COUNTEREXAMPLES TO THE GLUSHKOV MFN CONJECTURES

Abstract. We refute the Glushkov parts of Conjectures 1 and 2 of Berglund, van der Merwe, and le Roux. For every $k \geq 2$, we give a regular expression for which the Minimum Feedback Node (MFN) set of the Glushkov automaton has size k, whereas the unique minimum memoization set is a singleton outside MFN. We characterize all memoization sets eliminating infinite degree of ambiguity and show that every ordering of IAR^+ returns the entire MFN set. Both output-to-optimum ratios therefore equal k and are unbounded. The corresponding Thompson questions are not addressed.

1. The counterexample

Berglund, van der Merwe, and le Roux conjectured that, for every regular expression, a minimum memoization set can be chosen inside the Minimum Feedback Node (MFN) set, for both the Thompson and Glushkov constructions [1, Conjecture 1]. They equivalently conjectured that some ordering makes IAR^+ return a minimum memoization set [1, Conjecture 2 and Observation 3]. We disprove both Glushkov assertions.

We use the conventions of [1], in particular that $+$ is a primitive regular-expression operator. For an NFA A and $M \subseteq Q(A)$, write $A[M]$ for the corresponding memoized NFA. By the IDA characterization in [1, Theorem 1], $A[M]$ has infinite degree of ambiguity (IDA) if and only if there are distinct states p, q and a nonempty word v such that

$$p \xrightarrow{v} p, \qquad p \xrightarrow{v} q, \qquad q \xrightarrow{v} q,$$

with the q-cycle containing no state of M. Define

$$\mu(A) = \min\{|M| : M \subseteq Q(A),\ A[M] \text{ has no IDA}\}.$$

For an ordering $\prec$ of $\mathrm{MFN}(A)$, let $\mathrm{IAR}^+_\prec(A)$ denote the output of the procedure in [1, Definition 5].

For $k \geq 2$, let

$$R_k = \Big(\big(\underbrace{a^+ \mid \cdots \mid a^+}_{k \text{ occurrences}}\big)b\Big)^*, \qquad A_k = \mathrm{Gl}(R_k).$$

The alternatives are distinct syntax-tree occurrences and hence distinct positions in the Glushkov linearization; no algebraic simplification is applied before the construction. Write a_i^+ for the ith alternative. Let $x_1, \ldots, x_k$ be the positions labelled a, let y be the position labelled b, and put

$$X_k = \{x_1, \ldots, x_k\}.$$

Besides the initial state q_0, the nonempty transitions are

$$\delta(q_0, a) = X_k, \qquad \delta(x_i, a) = \{x_i\}, \qquad \delta(x_i, b) = \{y\}, \qquad \delta(y, a) = X_k$$

for $1 \le i \le k$. The final states are q_0 and y, and all states are useful.

Theorem 1. *For every $k \ge 2$ and every $M \subseteq Q(A_k)$,*

$$A_k[M] \text{ has no IDA} \quad \Longleftrightarrow \quad y \in M \quad \text{or} \quad X_k \subseteq M.$$

Moreover,

$$\mathrm{MFN}(A_k) = X_k, \qquad \mu(A_k) = 1,$$

and $\{y\}$ is the unique minimum memoization set. For every ordering $\prec$ of X_k,

$$\mathrm{IAR}^+_\prec(A_k) = X_k.$$

Proof. Algorithm 2 of [1] gives

$$a_i^+ \longmapsto (\varnothing, \{x_i\}, \mathrm{false}), \qquad (a_1^+ \mid \cdots \mid a_k^+) \longmapsto (\varnothing, X_k, \mathrm{false}).$$

Concatenation with b preserves the latter tuple, and the outer star changes only its nullability. Thus the root tuple is

$$(R_{\mathrm{op}}, R_{\mathrm{req}}, \mathrm{nullable}(R_k)) = (\varnothing, X_k, \mathrm{true}),$$

so

$$\mathrm{MFN}(A_k) = X_k.$$

Independently, every feedback vertex set contains all of X_k, since each x_i has a self-loop, while deleting X_k leaves no directed cycle.

Suppose first that $y \in M$. Every directed cycle avoiding M is then a self-loop at some x_i, and every word labelling such a cycle is a^r for some $r \ge 1$. If a state p returns to itself on a^r, then $p = x_j$ for some j, and

$$\delta(x_j, a^r) = \{x_j\}.$$

Thus

$$p \xrightarrow{a^r} q \qquad \text{and} \qquad q \xrightarrow{a^r} q$$

force $p = q$, so the IDA criterion cannot hold. If instead $X_k \subseteq M$, every directed cycle meets M, and again there is no IDA.

Conversely, suppose that

$$y \notin M \qquad \text{and} \qquad X_k \nsubseteq M.$$

Choose $x_j \in X_k \setminus M$ and an index $\ell \ne j$. With

$$p = x_\ell, \qquad q = x_j, \qquad v = ba,$$

we have

$$x_\ell \xrightarrow{ba} x_\ell, \qquad x_\ell \xrightarrow{ba} x_j, \qquad x_j \xrightarrow{ba} x_j.$$

The q-cycle visits only x_j and y, neither of which lies in M. The IDA criterion imposes no avoidance condition on the other two paths, so x_ℓ may itself lie in M. Hence $A_k[M]$ has IDA. This proves the characterization.

The empty set therefore fails, while $\{y\}$ succeeds, so

$$\mu(A_k) = 1.$$

The characterization also shows that every successful singleton is $\{y\}$, proving uniqueness.

Finally, IAR^+ starts from X_k. Removing any x_i produces $X_k \setminus \{x_i\}$, which contains neither y nor all of X_k and therefore has IDA. Thus no state is removable, regardless of the ordering. $\square$

Corollary 2. *The Glushkov parts of Conjectures 1 and 2 in* [1] *are false. In fact, for every $k \geq 2$ and every ordering $\prec$,*

$$\frac{|\,\mathrm{MFN}(A_k)|}{\mu(A_k)} = \frac{|\,\mathrm{IAR}^+_{\prec}(A_k)|}{\mu(A_k)} = k.$$

Thus both gaps are unbounded.

Proof. The unique minimum memoization set is $\{y\}$, whereas

$$y \notin \mathrm{MFN}(A_k) = X_k.$$

Consequently, no minimum memoization set is contained in MFN. The theorem also shows that every ordering of IAR^+ returns X_k. The ratios follow from $|X_k| = k$ and $\mu(A_k) = 1$. $\square$

References

[1] M. Berglund, B. van der Merwe, and I. le Roux, *Selective memoization for efficient backtracking regular expression matching*, Electron. Proc. Theor. Comput. Sci. **446** (2026), 1–17. doi:10.4204/EPTCS.446.1.